SHAKTI PRASAD MAHAPATRO

Penetration Tester | Offensive Security Engineer | VAPT Specialist | Ethical Hacker

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PROFESSIONAL SUMMARY

Offensive security professional and Penetration Tester with hands-on expertise in VAPT (Vulnerability Assessment and Penetration Testing) across web applications, APIs, Active Directory, mobile, cloud and network infrastructure. CEH-certified, with 18+ delivered security projects and 20+ critical vulnerability discoveries, using exploitation workflows mapped to MITRE ATT&CK and PTES. Proficient in Metasploit, Burp Suite, Nmap, BloodHound, SQLMap and Kali Linux. Experience extends into GRC advisory (ISO 27001, SOC 2, HIPAA, GDPR, PCI DSS), security automation, and technical training delivered to 600+ students, combining deep technical testing skill with clear, client-ready reporting and communication.

CORE COMPETENCIES

Offensive Security: Penetration Testing, Vulnerability Assessment and Penetration Testing (VAPT), Adversary Simulation, Exploit Development, OSINT and Reconnaissance, Social Engineering

Application & Infrastructure Security: Web Application Security Testing, API Security Testing, Network Penetration Testing, Mobile Application Penetration Testing, OWASP Top 10, Active Directory Security

Cloud & Detection: Cloud Security Assessment (AWS/Azure), SIEM and Threat Detection, Incident Response, Vulnerability Management, MITRE ATT&CK

Governance, Risk & Compliance: ISO 27001, SOC 2, PCI DSS, HIPAA, GDPR, Risk Treatment, Compliance Advisory

TECHNICAL SKILLS

Penetration Testing & Exploitation: Metasploit, Burp Suite, SQLMap, Nikto, Gobuster, Hydra, John the Ripper, BeEF, Netcat

Reconnaissance & OSINT: Nmap, Masscan, Shodan, Maltego, theHarvester, OSINT Framework

Active Directory & Identity: BloodHound, CrackMapExec, Impacket, Mimikatz, Kerberoasting, Pass-the-Hash

Web, API & Mobile Security: OWASP Top 10, OWASP API Top 10, IDOR, SQLi, XSS, SSRF, JWT Attacks, MobSF, Frida

Cloud & SIEM: AWS/Azure Misconfiguration, IAM Abuse, CSPM, Zero Trust, Splunk, Wazuh, Sentinel, QRadar, Sigma, KQL

GRC & Compliance: ISO 27001, SOC 2, HIPAA, GDPR, PCI DSS, DPIA, Risk Treatment, Annex A

Operating Systems, Labs & Methodology: Kali Linux, Parrot OS, Windows AD Lab, MITRE ATT&CK, PTES, CVSS, Kill Chain

PROFESSIONAL EXPERIENCE

Security Engineer - Penetration Testing

Aexantis Technologies Pvt. Ltd., Ghaziabad, UP | January 2026 – Present

- Deliver web application, API and network penetration testing engagements for enterprise and government clients, from scoping through exploitation to boardroom-ready reporting.
- Led a critical VAPT engagement that uncovered a chained authentication-bypass and privilege-escalation path in a client-facing production application, closed before exploitation risk materialised.
- Designed AI-assisted threat detection pipelines and security automation workflows for managed SOC engagements.
- Built the organisation's GRC practice from the ground up, including ISO 27001 ISMS rollout, SOC 2 and PCI DSS control mapping, and HIPAA/GDPR advisory for regulated enterprise clients.
- Led CERT-In empanelment strategy and GeM vendor registration to unlock government VAPT and cybersecurity contracts, sustaining 97% client retention across engagements.

Senior Executive – Cybersecurity & IT Compliance

Anmol Industries Limited, Kolkata, WB | July 2025 – November 2025

- Performed full-scope VAPT across web applications, APIs, industrial OT networks and internal infrastructure.
- Discovered and documented 20+ critical vulnerabilities, including authentication bypass and privilege escalation.
- Delivered CVSS-scored reports with proof-of-concept evidence and executive-level remediation roadmaps.

IT Specialist – Cybersecurity (SOC Support)

SBI General Insurance, Odisha | December 2024 – July 2025

- Conducted phishing simulations and email spoofing tests, reducing real phishing incidents by 60%.
- Implemented and validated SPF, DKIM and DMARC email authentication policies.
- Triaged 100+ security incidents across BFSI infrastructure.

Cybersecurity Trainer (Freelance)

IBM ICT Initiative, via Secuneus Tech

- Delivered offensive security training covering Nmap, Wireshark, Kali Linux and Metasploit to 600+ students.

Cybersecurity Analyst Intern

Tata Consultancy Services (TCS) | December 2023 – June 2024

- Analysed 5,000+ daily network events via Splunk and QRadar, identifying intrusions, brute-force attempts and lateral movement.
- Ran vulnerability scans using Nessus and OpenVAS, identifying 20+ high-risk issues and supporting remediation.

KEY OFFENSIVE SECURITY PROJECTS

Web Application VAPT – OWASP Top 10 Assessment

End-to-end black-box and grey-box penetration test covering authentication flaws, access-control gaps, SQL injection, XSS, SSRF and IDOR across a multi-tenant SaaS platform. Delivered PoC evidence with a CVSS-scored remediation report. Tools: Burp Suite, OWASP ZAP, SQLMap, Nikto.

Web Application Penetration Testing – E-Commerce & Fintech Platforms

Conducted full-scope web application penetration tests across e-commerce and fintech platforms, identifying payment-flow abuse, business logic flaws and insecure direct object references. Delivered client-ready reports with severity ratings and fix validation. Tools: Burp Suite Pro, OWASP Top 10, Postman, Manual Exploitation.

Authentication & Session Management Security Testing

Assessed login, session and token-handling flows for broken authentication, session fixation, weak password reset logic and MFA bypass paths. Provided remediation guidance aligned with OWASP ASVS. Tools: Burp Suite, OWASP ASVS, JWT Analyzer, Hydra.

Active Directory Security – AD Privilege Escalation Lab

Simulated common AD attack paths including misconfigured permissions, Kerberoasting, AS-REP Roasting, weak password hygiene and privilege escalation chains. Delivered identity hardening controls and attack-path visibility using BloodHound attack graphs. Tools: BloodHound, CrackMapExec, Impacket, John the Ripper.

Attack Simulation Pipeline

Designed and executed a full attack simulation workflow: disciplined scoping, OSINT reconnaissance, vulnerability validation, exploit execution and post-exploitation documentation, with remediation reporting mapped to the MITRE ATT&CK kill chain. Tools: PTES, MITRE ATT&CK, OSINT, Risk Report.

Cloud & IAM Misconfiguration Audit

Reviewed identity permissions, storage exposure, logging gaps and policy weaknesses across AWS/Azure environments against secure cloud architecture principles. Translated technical findings into board-level risk language and a prioritised remediation roadmap. Tools: IAM, CSPM, Zero Trust, Policy Review.

Android Mobile Application Security Review

Assessed mobile app attack surface including insecure local storage, cleartext API traffic, hardcoded secrets, weak certificate validation and platform permission misuse, combining static and dynamic testing. Tools: MobSF, Frida, ADB, Burp Suite.

REST API Authorization & Security Testing

Tested API endpoints for broken object-level authorization (BOLA/IDOR), JWT token weaknesses, excessive data exposure and rate-limit bypass across RESTful and GraphQL APIs. Delivered findings with reproduction steps and remediation guidance for developers. Tools: Postman, Burp Suite, JWT Analyzer, OWASP API Top 10.

CERTIFICATIONS

- CEH – Certified Ethical Hacker, EC-Council
- GCIA – GIAC Certified Intrusion Analyst
- ISO/IEC 27001 Information Security Associate
- Google Cybersecurity Professional Certificate
- Gemini AI Certified – Google
- OCI Generative AI Professional – Oracle
- TryHackMe – Junior Penetration Tester Path
- Diploma in Cybersecurity – Alison

EDUCATION

MBA – IT & Finance (Specialisation: Cybersecurity & Security Infrastructure) | Manipal University | August 2025 – Present

B.Sc. Computer Science | Berhampur University | December 2020 – July 2024

Intermediate (Class XII) – Physics, Chemistry & Mathematics | KSUB College of Teacher Education, Bhanjanagar

Matriculation (Class X) | St. Xavier's High School